REGULAR AND KOSZUL-REGULAR IMMERSIONS ARE PERFECT MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17813-17817 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17819-17831, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The cited input of the perfect-permanence card L16117 collected last wave; sequencing closes that internal edge.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Divisors, lemma regular-quasi-regular-immersion (regular sequences are Koszul-regular); the definition of perfect ring map via finite free presentations (chapter definition section); the Koszul complex exactness property defining Koszul-regular sequences.

Source reference. Original lemma 063K: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17831–17831. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A regular immersion of schemes is perfect. A Koszul-regular immersion of schemes is perfect.*

Proof. Since a regular immersion is a Koszul-regular immersion, see Divisors, Lemma 063K, it suffices to prove the second statement. This translates into the following algebraic statement: Suppose that $I \subset A$ is an ideal generated by a Koszul-regular sequence $f_1, \ldots, f_r$ of A. Then $A \to A/I$ is a perfect ring map. Since $A \to A/I$ is surjective this is a presentation of A/I by a polynomial algebra over A. Hence it suffices to see that A/I is pseudo-coherent as an A-module and has finite tor dimension. By definition of a Koszul sequence the Koszul complex $K(A, f_1, \ldots, f_r)$ is a finite free resolution of A/I. Hence A/I is a perfect complex of A-modules and we win. $\square$

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.